PCB fabrication specification and handover

USBGE-FAB-001 | Document revision 04 | 2026-10-16

Design baseline USBGE-A-WT-20261016-R04 | A (native schematic title revision A0)

Current fabrication data

Use this Rev 04 file manifest and the agreed FAB-005 build only after release approval. B_Fabrication/GerberX2 contains the complete14-file X2-compatible handoff:11 raw native X2 files and3 native RS274X files with metadata-only wrapping. No imaging command was changed. Native RS274X originals are separately supplied. FAB-006 traces every file to its source hash.

Four copper, both solder masks, both paste layers (bottom paste intentionally empty: no bottom SMT), both legends, profile, mechanical1 and PTH/NPTH drill artwork are present. Actual drill programs are retained Rev 03 native NC and explicit-decimal PTH104/NPTH2, all 106 matching current pad/via geometry within0.00011 mm.

Origin / polarity / data interpretation

All selected artwork is in the relative lower-left board frame,70x34 mm with R4 corners. Shared coordinate metadata is bound by the assembly script; mask flashes independently align to every applicable native pad centre. The three wrappers add X2 file-function/polarity/common-frame metadata only; raw native RS274X is authoritative for those imaging commands.

Internal planes are NEGATIVE polarity; interpret with the board profile. ODB++ matrix explicitly identifies both GND planes as negative. Do not invert them or assume a plane Gerber is a positive copper image. Retained Rev 03 native IPC2581 revisionB XML and ODB++ with CAD netlist provide alternative integrated manufacturing data.

Build and CAM review

FAB-005 is a proposed complete construction, not a purchase approval. Current digital edge minimum is 0.250933 mm, supporting a proposed 0.25 mm requirement; the earlier0.254 mm proposal is not demonstrated. Agree actual finished stack/finish/tolerances/impedance and effective edge-rule limit. Supplier CAM must review all layers, drill registration, negative planes, electrical-test netlist, coupons and panel tooling before production.

C5 is approved and implemented as 22 uF, 10 V, X5R, 0603, Murata GRM188R61A226ME15D. The CAP_0603 footprint is retained. Native schematic parameters and PCB comment/imported parameters agree; all unrelated native streams are unchanged. Fabrication geometry is unchanged and Rev 03 artwork is carried forward by source-stream equivalence; native metadata-bearing exports remain pending.